[Web Serial Telemetry] [Google Gemini 2.5 Flash] [Lead-I ECG at 100 Hz] [13 Indian Vernacular Languages]

Bridging the Last Mile in Rural Healthcare with Multimodal AI and Edge Medical Hardware

A low-cost, browser-native clinical workstation empowering community health workers (ASHA/ANM) with vernacular voice intake, paper report vision OCR, and real-time streaming ECG & SpO2 hardware telemetry.

1 **Multimodal Intake**
Vernacular Voice, OCR & Vitals

→

2 **Web Serial Edge Gateway**
Zero-Driver 100Hz Ingestion

→

3 **Dual AI & Safety Core**
Gemini 2.5 + Deterministic Red-Flags

→

4 **Tele-Doctor Review**
15s ECG Calipers & Printable Dossier

DEVELOPER & LEAD
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HARDWARE ARCHITECTURE
ESP32 + AD8232 ECG + MAX30102 SpO2
Hardware BOM Cost: < $10 (₹800 INR)

The Frontline Diagnostic Vacuum at Primary Health Centres

Over 800 million citizens in rural India rely on overburdened Primary Health Centres (PHCs). Frontline health workers have no diagnostic equipment, causing severe delays in identifying acute emergencies.

CRITICAL BREAKDOWN IN THE CURRENT RURAL TRIAGE CHAIN:

Frontline ASHA Visit
Zero diagnostic sensors available

→

Vernacular Barrier
Dialects & paper reports untranslated

→

4+ Hour Delay
Unrecognized cardiac / hypoxia crises

→

Severe Complications
Irreversible tissue & cardiac damage

70% vs 30%

DOCTOR DISTRIBUTION MISMATCH

70% of population in rural India; only 30% of doctors practice outside cities.

$500+

TRADITIONAL MEDICAL DEVICE COST

Hospital monitors are unaffordable and require specialized trained technicians.

22+ Dialects

REGIONAL LANGUAGE CHASM

Patients cannot express complex symptom histories in standard medical English.

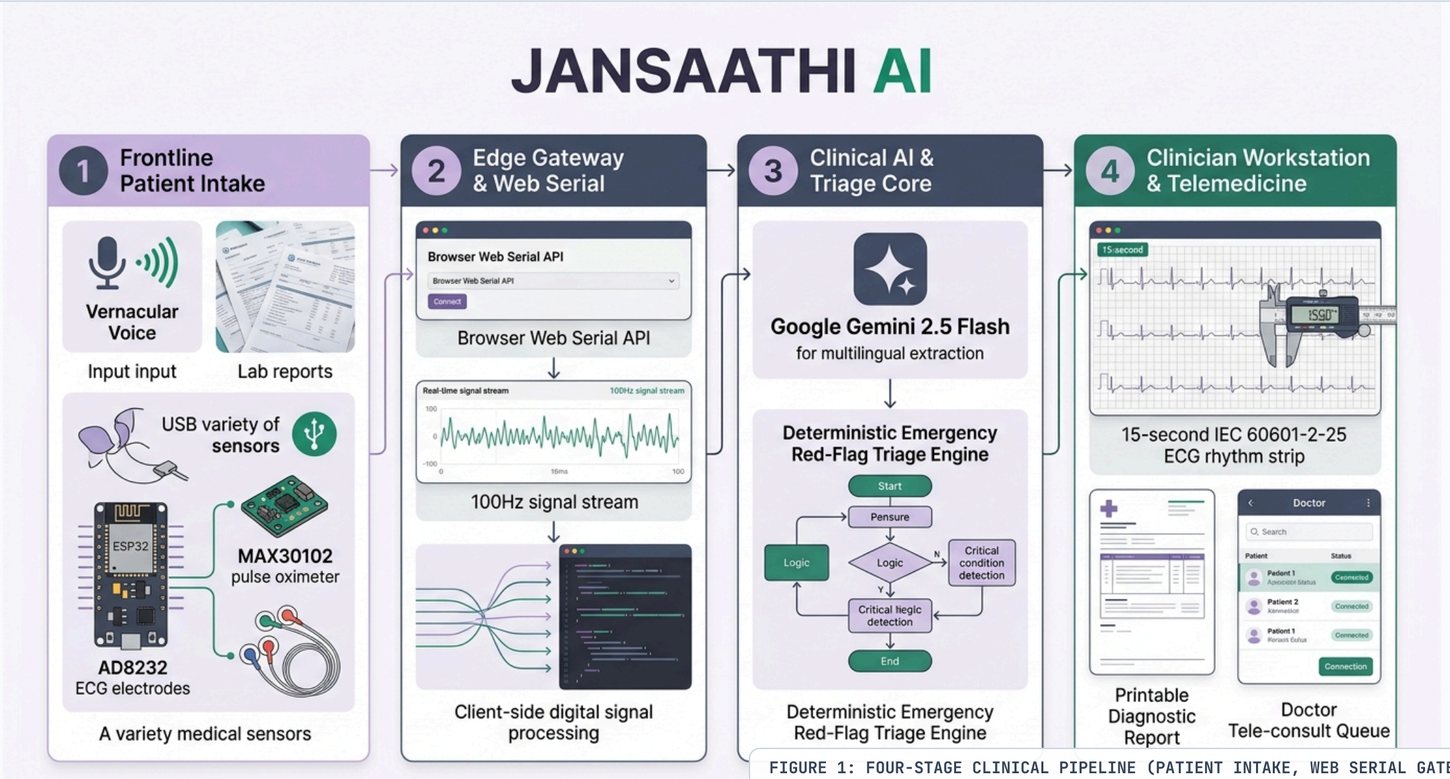


FIGURE 1: FOUR-STAGE CLINICAL PIPELINE (PATIENT INTAKE, WEB SERIAL GATEWAY, GEMINI AI ENGINE, DOCTOR TELE-CONSULT)

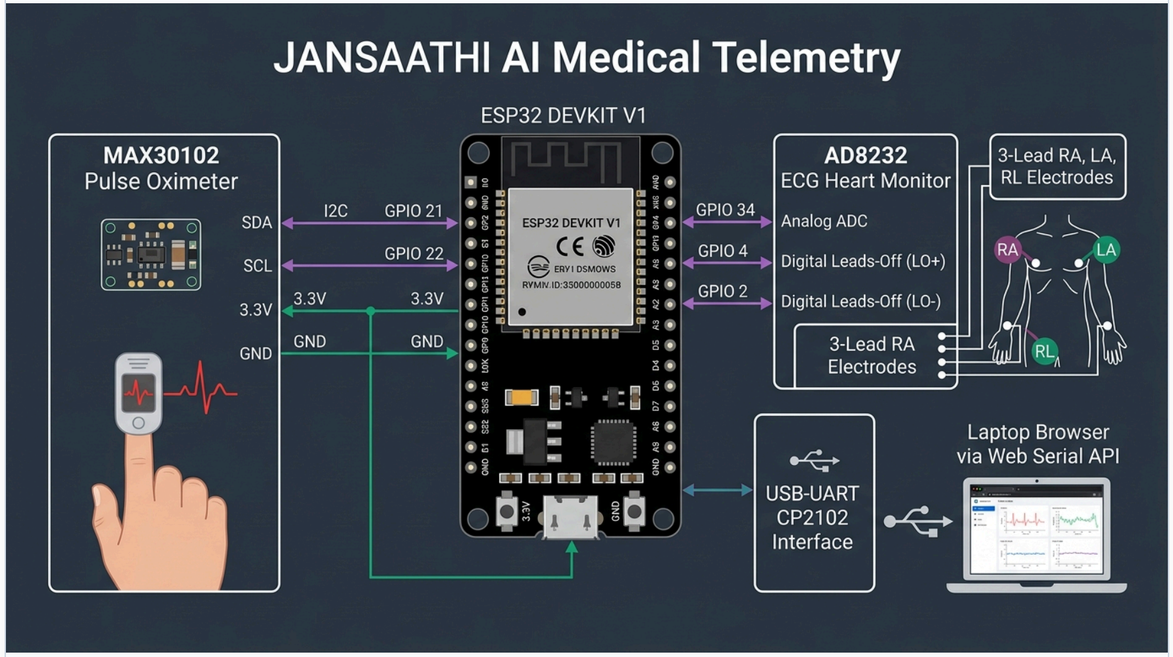


FIGURE 2: ESP32 PINOUT MAPPING AND MEDICAL SENSOR INTERCONNECT

Sub-$10 Medical Hardware with Zero Driver Installation

Uses standard Web Serial API (`navigator.serial`) at 115200 baud. The browser directly interfaces with the ESP32 microcontroller, reading optical and electrophysiological signals.

MAX30102 Pulse Oximeter (I2C)

SDA: GPIO 21 • SCL: GPIO 22

Digital optical AC tracking filter (`irDc = (irDc * 255 + ir)/256`) eliminates ambient light drift, extracting SpO2, Pulse Rate, and arterial plethysmograph (PPG) at 50 Hz.

AD8232 Lead-I ECG (ADC + Digital LO)

ADC: GPIO 34 • LO±: GPIO 4 & 2

Lead-I RA/LA/RL electrodes sampled at 100 Hz. Features 25-sample noise debounce and analog rail voltage check (ADC < 120 or > 3980) to eliminate lead fluttering.

Web Serial Gateway Engine

Auto-Baud 115200

Parses JSON bursts (`ecg_samples`, `ppg_samples`) and ASCII key-value lines with automated port reconnection and COM port contention recovery.

Dual-Tier Intelligence: Gemini 2.5 Flash + Deterministic Red-Flags

Clinical triage cannot rely purely on generative probabilistic models. JANSAATHI AI implements a strict clinical safety net where acute emergencies are governed by deterministic rules.

Tier 1: Google Gemini 2.5 Flash Extraction

Converts vernacular voice transcripts and paper lab report images into structured, FHIR-compliant clinical JSON with symptom timelines and missing information queries.

Tier 2: Deterministic Rule Safety Net

Zero hallucination tolerance: `RULE_CHEST_PAIN`, `RULE_RESPIRATORY_DISTRESS`, `RULE_CRITICAL_SPO2` (<90%), and `RULE_ARRHYTHMIA` instantly trigger `URGENT REVIEW`.

Tier 3: Tele-ICU & PHC Doctor Escalation

Automated case prioritization in doctor's queue with one-click disposition (Admit, Tele-Refer, Follow-up) and complete tamper-evident clinical audit trails.

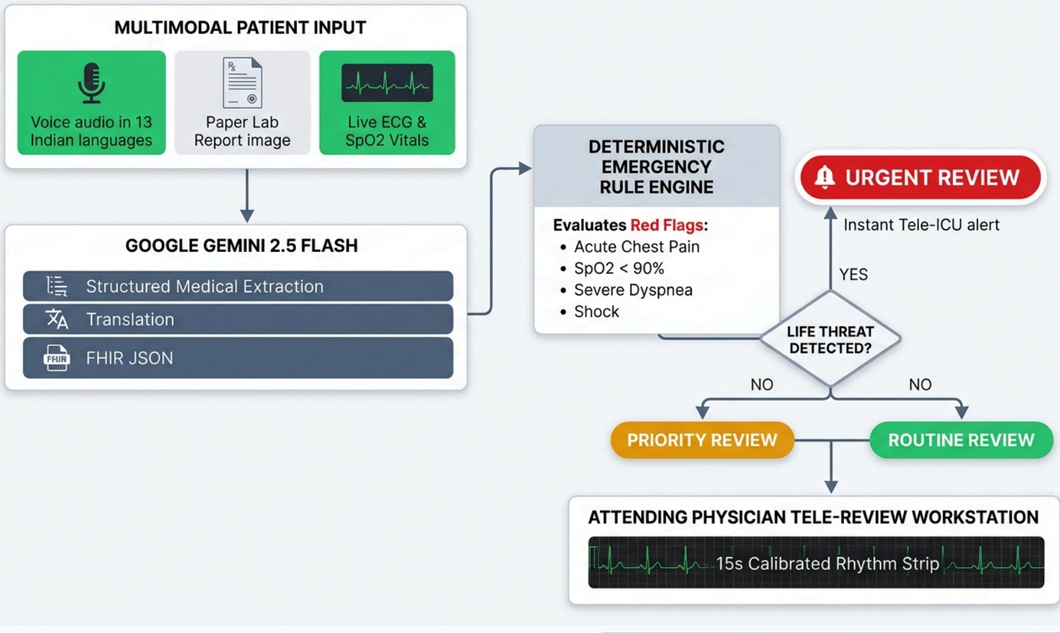


FIGURE 3: DETERMINISTIC EMERGENCY DECISION TREE

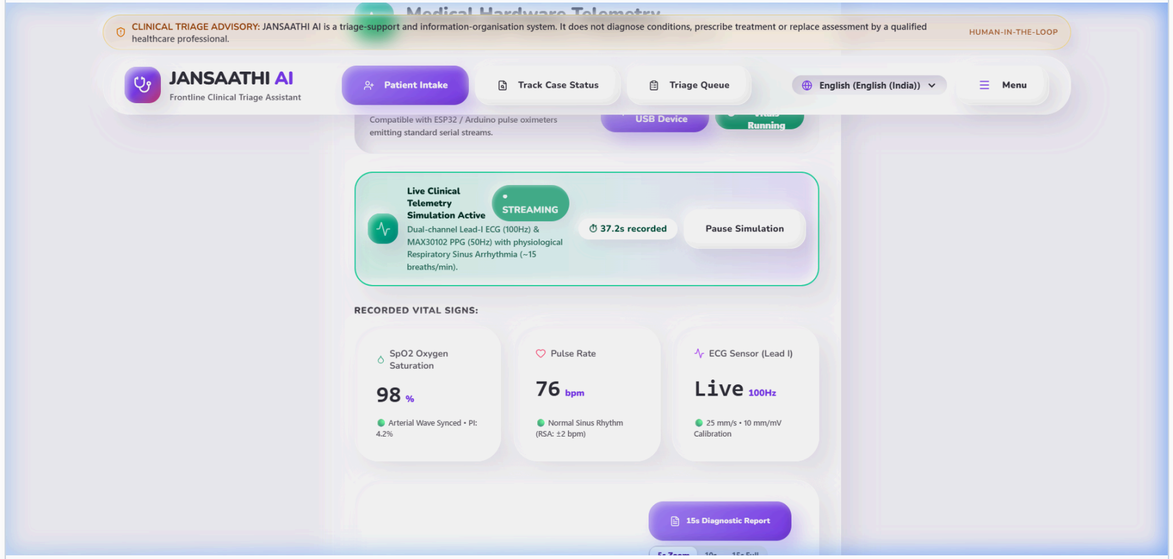


FIGURE 4: LIVE PATIENT MONITOR WITH RESPIRATORY SINUS ARRHYTHMIA

Real-Time Telemetry with Respiratory Sinus Arrhythmia (RSA)

The system renders genuine, active physiological telemetry. Both physical USB sensors and the built-in demo simulator feature dynamic physiological variability.

Respiratory Sinus Arrhythmia (RSA)

72 – 76 bpm natural undulation

Resting heart rate fluctuates in lockstep with a 4.0-second respiration cycle (15 breaths/min), accompanied by subtle biological micro-jitter (±0.8 bpm).

Synchronized Arterial PPG Pulse Wave

~180ms Pulse Transit Time

The photoplethysmograph waveform is coupled to the ECG R-peak with physiological ventricular-to-peripheral transit delay, displaying steep anacrotic upstroke and aortic dicrotic notch.

Interactive Caliper Freeze

Caliper Inspection

Clinicians can freeze live scrolling waveforms on demand to inspect caliper measurements and resume streaming seamlessly without losing connection state.

Hospital-Grade Cardiogram Strip with Electrophysiological Metrics

Standard 15.0-second Lead-I acquisition rendered at standard 25 mm/s paper speed and 10 mm/mV voltage calibration on medical millimeter grid.

MEAN R-R INTERVAL
811 ms
Dynamic heart rate coupling

PR INTERVAL
152 ms
Atrioventricular conduction

QRS DURATION
88 ms
Normal narrow complex

BAZETT QTC
418 ms
< 440 ms normal cutoff

Printable PDF Diagnostic Dossier: One-click generation of doctor-signed A4 clinical reports containing patient demographics, full 15s Lead-I rhythm strip, SpO2 PPG waveform, and diagnostic interpretation.



FIGURE 5: ONE-CLICK PRINTABLE A4 DIAGNOSTIC DOSSIER

Working at the Grand Finale

Live Plug-and-Play USB Vitals Station
Real-time Lead-I ECG (100Hz) & MAX30102 pulse oximetry streaming via Web Serial.

Multimodal Vernacular Voice & Vision OCR
Google Gemini 2.5 Flash intake across major Indian regional languages with lab report extraction.

15s Diagnostic Strip & Calipers
IEC 60601-2-25 calibration, live waveform freeze, and printable doctor PDF dossier.

Rigorous Quality Assurance & Benchmarks

21 / 21
VITEST TESTS PASSING

< 40 ms
INGESTION LATENCY

0 Errors
STRICT TYPESCRIPT

< $10
HARDWARE BOM COST

Repository & Presenter Details:
- GitHub: `https://github.com/b2bhau570/jansaathi-ai`
- Lead Developer: **Satyabrata Pradhan** (b2bhau570)